

# Module 09: Multi-Gaussian Expansion & Linear Light Profiles

Theory Companion — Learning to Autolens

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## Abstract

This document develops the mathematical framework for linear light profile fitting and Multi-Gaussian Expansions (MGEs) as used in PyAutoLens. We show that solving for light profile intensities is a linear algebra problem identical in structure to the pixelized source inversion of Module 05, derive the maximum-likelihood and NNLS solutions, and explain why Gaussians form a particularly efficient basis. References: Emsellem, Monnet & Bacon (1994); Cappellari (2002); Lawson & Hanson (1974).

## Contents

# 1 Linear Algebra of Light Profile Fitting

In a traditional lens model every light profile has an **intensity** parameter that the non-linear search must explore. A *linear light profile* removes intensity from the search: given fixed shape parameters, the optimal intensity is found by solving a linear system.

## 1.1 The Forward Model

Suppose we have  $K$  light profile components (e.g. bulge, disk, lensed arcs), each with a known 2D shape but unknown intensity  $a_k$ . Let  $\mathbf{b}_k$  be the PSF-convolved image of component  $k$  evaluated at unit intensity — a column vector of  $N_d$  image pixels. Stack these into the **blurring matrix**  $\mathbf{B} = [\mathbf{b}_1 \mid \cdots \mid \mathbf{b}_K]$ , which has dimensions  $N_d \times K$ . The model image is then:

### Forward Model

$$\mathbf{d} = \mathbf{B} \mathbf{a} + \mathbf{n}, \quad (1)$$

where  $\mathbf{a} = (a_1, \dots, a_K)^T$  is the intensity vector and  $\mathbf{n}$  is the noise with covariance  $\mathbf{C} = \text{diag}(\sigma_1^2, \dots, \sigma_{N_d}^2)$ .

This is *exactly* the same structure as the pixelized source inversion from Module 05 (??  $\leftrightarrow$  Module 05 eq. 1), with the replacement

$$\underbrace{\mathbf{M}}_{\text{lensing+PSF}} \mathbf{s} \longleftrightarrow \underbrace{\mathbf{B}}_{\text{PSF-convolved basis}} \mathbf{a}.$$

The source pixels  $\mathbf{s}$  are replaced by intensities  $\mathbf{a}$ , and the lensing operator by the basis image matrix. All the linear algebra carries over.

## 1.2 Maximum-Likelihood Solution

Minimizing  $\chi^2(\mathbf{a}) = (\mathbf{d} - \mathbf{B}\mathbf{a})^T \mathbf{C}^{-1} (\mathbf{d} - \mathbf{B}\mathbf{a})$  and setting  $\partial\chi^2/\partial\mathbf{a} = 0$  gives the weighted least-squares solution:

### ML Intensity Solution

$$\hat{\mathbf{a}}_{\text{ML}} = (\mathbf{B}^T \mathbf{C}^{-1} \mathbf{B})^{-1} \mathbf{B}^T \mathbf{C}^{-1} \mathbf{d} \quad (2)$$

The matrix  $\mathbf{B}^T \mathbf{C}^{-1} \mathbf{B}$  is only  $K \times K$  (typically  $K \sim 20\text{--}60$  for an MGE), so the solve is trivially cheap. Unlike the pixelized source case, *no regularization is needed* because  $K \ll N_d$  and the system is well-conditioned.

# 2 Positive-Only Constraint (NNLS)

The unconstrained solution (??) can yield negative  $a_k$ . While mathematically valid, a negative surface brightness is unphysical for a stellar light component.

PyAutoLens therefore solves the **Non-Negative Least Squares** (NNLS) problem by default:

$$\hat{\mathbf{a}}_{\text{NNLS}} = \arg \min_{\mathbf{a} \geq 0} \|\mathbf{C}^{-1/2} (\mathbf{d} - \mathbf{B}\mathbf{a})\|^2 \quad (3)$$

The standard algorithm is that of Lawson & Hanson (1974): an active-set method that iteratively partitions the indices into a “free” set (non-zero coefficients) and a “fixed” set (zeroed coefficients), solving the unconstrained sub-problem on the free set at each step. Convergence is guaranteed in a finite number of iterations.

### Exercise

Construct a toy 1D example with three Gaussian basis functions where the unconstrained least-squares solution gives one negative amplitude. Verify that NNLS sets it to zero and redistributes flux to the remaining components.

## 3 Multi-Gaussian Expansion Theory

### 3.1 The MGE Ansatz

A Multi-Gaussian Expansion represents a surface brightness profile as a sum of 2D Gaussians:

$$I(R) = \sum_{j=1}^K a_j \exp\left(-\frac{R^2}{2\sigma_j^2}\right) \quad (4)$$

where  $R$  is the elliptical radius appropriate to each component and  $\{\sigma_j\}$  are the Gaussian widths (dispersions).

Historically, MGEs were developed for stellar dynamical modeling (Emsellem, Monnet & Bacon 1994) and efficient fitting algorithms were given by Cappellari (2002). Their adoption in lens modeling is more recent.

### 3.2 Why Gaussians Are Special

Three properties make Gaussians a uniquely convenient basis:

1. **Analytic PSF convolution.** If the PSF is itself approximated by a Gaussian with dispersion  $\sigma_p$ , the convolution of a basis Gaussian ( $\sigma_j$ ) with the PSF is another Gaussian:

#### Gaussian Convolution Property

$$\sigma_{\text{eff}} = \sqrt{\sigma_j^2 + \sigma_p^2} \quad (5)$$

The convolved image is a Gaussian with the same centre and ellipticity, but broadened dispersion  $\sigma_{\text{eff}}$ .

In practice, PyAutoLens convolves numerically with an arbitrary PSF kernel (not necessarily Gaussian), but the near-Gaussian shape of each basis image means the convolution is well-behaved.

2. **Analytic deprojection.** A 2D Gaussian deprojects to a 3D Gaussian (with the same  $\sigma$  along the projected axis), enabling analytic computation of the gravitational potential and forces — critical for dynamical modeling (Emsellem+94 Sec. 3).
3. **Computational speed.** Evaluating  $\exp(-R^2/2\sigma^2)$  is fast and numerically stable; no special functions (incomplete gamma, hypergeometric) are needed, unlike Sérsic profiles.

## 4 MGE as a Basis Expansion

### 4.1 The Sigma Grid

PyAutoLens constructs the Gaussian widths on a logarithmic grid spanning from sub-pixel scales to the mask radius:

$$\sigma_j \in \{\sigma_{\min}, \sigma_{\min} \cdot r, \sigma_{\min} \cdot r^2, \dots, \sigma_{\max}\}$$

where  $r$  is the common ratio. Typical values:  $\sigma_{\min} \approx 0.01''$  (sub-pixel),  $\sigma_{\max} \approx$  mask radius, and  $K \sim 20\text{--}30$  Gaussians per group. The log-spacing ensures roughly equal sensitivity per octave of angular scale.

### 4.2 Groups and gaussian\_per\_basis

For multi-component morphologies (e.g. bulge + disk, or lens galaxy + lensed arcs), the Gaussians are divided into **groups**. Each group shares:

- a common centre (**centre**),
- common ellipticity components (**ell\_comps**).

Within each group, the  $K_g$  Gaussian widths  $\{\sigma_j\}$  are fixed (non-linear parameters would be needed only to adjust the grid, which is not done), and the  $K_g$  intensities  $\{a_j\}$  are solved linearly. The only non-linear parameters per group are:

$$\underbrace{x_c, y_c}_{\text{centre}} + \underbrace{e_1, e_2}_{\text{ellipticity}} = 4 \text{ parameters/group.}$$

#### PyAutoLens Implementation

A two-group MGE (lens light + source light) is set up as:

```
# Group 1: lens galaxy (30 Gaussians)
lens_bulge = af.Model(
    al.lp_linear.Gaussian,
    sigma=af.LogUniformPrior(
        lower_limit=0.01, upper_limit=10.0
    ),
)
# Group 2: source galaxy (20 Gaussians)
source = af.Model(
    al.lp_linear.Gaussian,
    sigma=af.LogUniformPrior(
        lower_limit=0.01, upper_limit=5.0
    ),
)
```

In practice, the `al.util.model.mge_model_from()` utility constructs the full model automatically (see Sec. ??).

## 5 Parameter Space Reduction

The central advantage of MGE + linear solving is the dramatic reduction in the dimensionality of the non-linear search.

The trade-off:

Table 1: Non-linear parameter count comparison for a bulge + disk lens light model.

| Approach                                                           | Non-linear params | Linear params |
|--------------------------------------------------------------------|-------------------|---------------|
| Sérsic bulge + Sérsic disk<br>(each: $x, y, e_1, e_2, R_e, n, I$ ) | 14                | 0             |
| MGE 2-group (30+30 Gaussians)<br>(each group: $x, y, e_1, e_2$ )   | $2 \times 4 = 8$  | 60            |
| MGE 1-group (shared geometry)                                      | 4                 | 30            |

- **Fewer non-linear dimensions**  $\Rightarrow$  the sampler converges faster and explores the posterior more efficiently.
- **Each likelihood evaluation is more expensive:** computing  $K$  basis images and solving the  $K \times K$  linear system adds overhead compared to evaluating a single Sérsic profile.

In practice the dimensionality reduction dominates, and MGE models typically complete 5–10 $\times$  faster than equivalent Sérsic models.

#### The mge\_model\_from Utility

```
import autolens as al
from autolens.util.model import mge_model_from

# Build an MGE model for the lens light
lens_bulge = mge_model_from(
    model=al.lp_linear.Gaussian,
    gaussian_per_basis=30,
)
# Returns an af.Collection of 30 Gaussian
# models sharing centre and ell_comps,
# with log-spaced sigma values.
```

## 6 MGE in the SLAM Pipeline

The standard SLAM (Source, Light, Mass) pipeline has four stages when using pixelized sources (Module 04). When MGE linear profiles are used for *both* lens light and source light, the pipeline simplifies to three stages:

1. **SOURCE LP** — Fit a parametric (MGE) source with a simple mass model (SIE + shear). The source MGE intensities are solved linearly; the non-linear search explores only mass parameters and source geometry (centre, ellipticity).
2. **LIGHT LP** — Fix the mass model from Stage 1 and fit the lens galaxy light with an MGE. This decouples the lens light from the mass, allowing a flexible multi-Gaussian representation without destabilizing the mass model.
3. **MASS TOTAL** — Fit the total mass model (e.g. power-law ellipsoid + shear) with lens light and source light initialised from previous stages.

**Why no SOURCE PIX stage?** The pixelized source stage exists to capture irregular source morphology that parametric profiles miss. When the source is already represented by an MGE with  $\sim 20$ – $30$  Gaussians, it has enough flexibility to represent most source morphologies without a

pixelized reconstruction. The SOURCE PIX stage is needed only when the source has structure on scales finer than the densest Gaussian in the grid (e.g. highly irregular merging galaxies).

## 6.1 Decision Guide

Table 2: When to use each source representation.

| Method       | Best for                                                                                                                      |
|--------------|-------------------------------------------------------------------------------------------------------------------------------|
| Sérsic       | Quick initial models; sources with known smooth morphology; when only 1–2 light components are needed.                        |
| MGE (linear) | Default choice for most lenses; multi-component morphologies; fast convergence when the non-linear parameter budget is tight. |
| Pixelized    | Highly irregular or merging sources; when residuals from parametric fits show systematic structure; substructure studies.     |

### Exercise

Take the SLAM pipeline from Module 04 and replace the SOURCE PIX stage with a SOURCE LP stage using an MGE source. Compare the Bayesian evidence of the two approaches on the same dataset.

## References

- Cappellari, M. (2002), MNRAS, 333, 400
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